

EXPLICIT DIRICHLET COMPOSITION–DIFFERENTIATION SHIFTS RESOLVE BOTH BEREZIN-SECTORIAL CONSTRUCTION ALTERNATIVES

CANDIDATE SOLUTION PACKET FOR ARXIV:2601.02817

ABSTRACT. The conclusion of Mahapatra–Mukherjee–Sen–Birbonshi–Paul asks whether scalar shifts of composition–differentiation operators on the Dirichlet space can systematically produce (i) Berezin sectorial but nonsectorial operators or (ii) operators that are sectorial in both senses with different indices. We give both constructions from one symbol. For $\phi(z) = z/2$ and $Df = f' \circ \phi$, the Berezin range is a disk of radius $b < 9/20$, whereas the numerical range contains the disk of radius $\sqrt{7/24} > 9/20$. Thus $aI + D$ is Berezin sectorial but not sectorial whenever $9/20 < a < \sqrt{7/24}$. Moreover, $I + D$ is sectorial and its classical index is strictly larger than its Berezin index. The only finite calculation in the proof is supplied as an exact rational Bernstein certificate.

Status. Candidate full answer, likely valid, subject to human review. The result answers both alternatives of the intended *shifted* composition–differentiation construction problem.

1. SOURCE PROBLEM AND SCOPE

The source is S. Mahapatra, S. Mukherjee, A. Sen, R. Birbonshi, and K. Paul, *An introduction of Berezin sectorial operators and its application to Berezin number inequalities*, arXiv:2601.02817 (2026) [1]. On PDF page 22 the authors explicitly motivate the question by “applying certain shifts to composition-differentiation operators.” The conclusion on PDF page 23 asks whether such operators can be systematically constructed “to be Berezin sectorial but not sectorial or both Berezin sectorial and sectorial but with different indices.”

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inequalities highlight the effectiveness of the Berezin sectorial framework in refining Berezin number estimates and understanding the geometry of the Berezin range. These results highlight that the inequalities obtained for Berezin sectorial operators cannot be achieved by standard sectorial operators alone. While the Berezin range of several operators on the Dirichlet space exhibits properties analogous to those in the Hardy-Hilbert space, it remains an open problem whether composition-differentiation operators can be systematically constructed to be Berezin sectorial but not sectorial or both Berezin sectorial and sectorial but with different indices. Addressing this question may lead to further insights into the interaction between Berezin range geometry and Berezin number inequalities.

DECLARATIONS

Acknowledgements. This work originated during the research visit of Dr. Anirban Sen to the Department of Mathematics at Jadavpur University, Kolkata, India. He gratefully acknowledges support from the Czech Science Foundation

Source evidence: conclusion, PDF page 23 of arXiv:2601.02817.

We answer both alternatives with scalar shifts of a single composition–differentiation operator, precisely in the sense of the preceding source paragraph. We do not claim that $aI + D_\phi$ itself can be rewritten as one unshifted D_ψ .

2. DEFINITIONS

We use the Dirichlet Hilbert space and norm fixed by the source:

$$\mathcal{D} = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \|f\|_{\mathcal{D}}^2 = \sum_{n \geq 0} (n+1) |a_n|^2 < \infty \right\}.$$

Thus

$$e_n(z) = \frac{z^n}{\sqrt{n+1}} \quad (n \geq 0)$$

is an orthonormal basis. The reproducing kernel at $\lambda \in \mathbb{D}$ and its squared norm are

$$k_\lambda(z) = \sum_{n \geq 0} \frac{\bar{\lambda}^n z^n}{n+1}, \quad K(t) := \|k_\lambda\|^2 = \sum_{n \geq 0} \frac{t^n}{n+1} = \frac{-\log(1-t)}{t}, \quad t = |\lambda|^2,$$

with the value at $t = 0$ understood by continuity.

For a bounded operator T on $\mathcal{D}$,

$$W(T) = \{\langle Tx, x \rangle : \|x\| = 1\}, \quad \text{Ber}(T) = \{\langle T\hat{k}_\lambda, \hat{k}_\lambda \rangle : \lambda \in \mathbb{D}\}.$$

For $0 \leq \theta < \pi/2$, put $S_\theta = \{z : |\arg z| \leq \theta\}$. The operator is sectorial if $W(T) \subseteq S_\theta$, and Berezin sectorial if $\text{Ber}(T) \subseteq S_\theta$, for some such θ . We denote the infima of admissible semi-angles by $\theta_{\text{num}}(T)$ and $\theta_{\text{Ber}}(T)$, respectively.

For the analytic self-map $\phi(z) = z/2$, let

$$D = D_\phi, \quad (Df)(z) = f'(\phi(z)) = f'(z/2).$$

3. IDEA OF THE PROOF

In the monomial basis, D is a compact backward weighted shift. This has two complementary consequences. First, its Berezin symbol can be summed radially: it is $\bar{\lambda}$ times a positive function of $|\lambda|^2$, so its Berezin range is exactly a centered disk. An exact polynomial majorant, certified in Bernstein form, puts that disk inside radius $9/20$. Second, the first two shift weights already produce, on a three-dimensional compression, a numerical circle of radius $\sqrt{7/24}$. The rational threshold $9/20$ lies strictly between the two radii. Translating both ranges by $a > 0$ then yields the desired sectorial separations. A uniform norm bound below one makes the shift by I classically sectorial as well, but with a strictly larger index.

4. THE CONSTRUCTION THEOREM

Define

$$F(t) = \sum_{n \geq 1} \frac{n}{n+1} \left(\frac{t}{2}\right)^{n-1}, \quad b = \max_{0 \leq t \leq 1} \frac{\sqrt{t} F(t)}{K(t)},$$

where the quotient is extended continuously by zero at $t = 1$.

Theorem 1. *For $Df = f'(z/2)$ on $\mathcal{D}$, the following hold.*

- (1) $\text{Ber}(D) = \{z : |z| \leq b\}$ and $b < 9/20$.
- (2) $W(D)$ contains $\{z : |z| \leq \sqrt{7/24}\}$.

(3) For every

$$\frac{9}{20} < a < \sqrt{\frac{7}{24}},$$

the shifted composition-differentiation operator $T_a = aI + D$ is Berezin sectorial but not sectorial.

(4) The operator $T_1 = I + D$ is both sectorial and Berezin sectorial, and its optimal indices satisfy

$$\theta_{\text{Ber}}(T_1) = \arcsin b < \arcsin \frac{9}{20} < \arcsin \sqrt{\frac{7}{24}} \leq \theta_{\text{num}}(T_1) < \frac{\pi}{2}.$$

Consequently one explicit symbol gives systematic families resolving both alternatives in the source problem.

Proof. Acting on the orthonormal basis gives

$$(1) \quad De_0 = 0, \quad De_n = \alpha_n e_{n-1}, \quad \alpha_n = n \sqrt{\frac{n}{n+1}} 2^{1-n} \quad (n \geq 1).$$

In particular $\alpha_n \rightarrow 0$, so D is compact and bounded.

Using the convention that the inner product is linear in its first variable, the kernel expansion and (1) yield

$$\langle Dk_\lambda, k_\lambda \rangle = \bar{\lambda} \sum_{n \geq 1} \frac{n}{n+1} \left(\frac{|\lambda|^2}{2} \right)^{n-1} = \bar{\lambda} F(t).$$

Hence

$$(2) \quad \tilde{D}(\lambda) = \bar{\lambda} \frac{F(t)}{K(t)}, \quad t = |\lambda|^2.$$

The radial modulus in (2) is continuous, vanishes at $t = 0$, and also tends to zero as $t \rightarrow 1^-$, because $F(1) < \infty$ while $K(t) \rightarrow \infty$. It therefore attains a maximum b . Every angle occurs, and the intermediate value theorem supplies every radius between zero and b . Thus $\text{Ber}(D) = b\overline{\mathbb{D}}$.

It remains to prove the strict rational bound for b . For $0 \leq t \leq 1$, set

$$P(t) = \sum_{n=0}^8 \frac{t^n}{n+1},$$

$$N(t) = \sum_{n=1}^8 \frac{n}{n+1} \left(\frac{t}{2} \right)^{n-1} + 2 \left(\frac{t}{2} \right)^8.$$

Plainly $K(t) \geq P(t)$. Also

$$\sum_{n \geq 9} \frac{n}{n+1} \left(\frac{t}{2} \right)^{n-1} \leq \frac{(t/2)^8}{1-t/2} \leq 2(t/2)^8,$$

so $F(t) \leq N(t)$. Consider the exact degree-17 polynomial

$$(3) \quad q(t) = \frac{81}{400} P(t)^2 - t N(t)^2.$$

Write $q(t) = \sum_{j=0}^{17} q_j t^j$. Its degree-32 Bernstein coefficients on $[0, 1]$ are

$$\beta_k = \sum_{j=0}^{\min(k, 17)} q_j \frac{\binom{k}{j}}{\binom{32}{j}}, \quad 0 \leq k \leq 32.$$

Exact rational expansion gives $\beta_k > 0$ for all 33 indices, with

$$\min_{0 \leq k \leq 32} \beta_k = \beta_{28} = \frac{55075026511}{201587097600000} > 0.$$

Because the Bernstein basis functions are nonnegative and sum to one, $q(t) > 0$ throughout $[0, 1]$. The attached checker performs this finite expansion using rational arithmetic only. From (3),

$$\frac{\sqrt{t}F(t)}{K(t)} \leq \frac{\sqrt{t}N(t)}{P(t)} < \frac{9}{20},$$

and therefore $b < 9/20$.

For the numerical lower bound, compress D to $E = \text{span}\{e_0, e_1, e_2\}$. The first two weights are

$$\alpha_1 = \frac{1}{\sqrt{2}}, \quad \alpha_2 = \sqrt{\frac{2}{3}}.$$

The largest eigenvalue of the real part of this three-by-three weighted shift is

$$r_0 = \frac{1}{2} \sqrt{\alpha_1^2 + \alpha_2^2} = \frac{1}{2} \sqrt{\frac{1}{2} + \frac{2}{3}} = \sqrt{\frac{7}{24}}.$$

Thus $r_0 \in W(D)$. If $U_\vartheta e_n = e^{in\vartheta} e_n$, then $U_\vartheta^* D U_\vartheta = e^{i\vartheta} D$. Unitary invariance gives $r_0 e^{i\vartheta} \in W(D)$ for every ϑ . Convexity of the numerical range (Toeplitz–Hausdorff; see [2]) now gives $r_0 \mathbb{D} \subseteq W(D)$. The two radii are strictly separated because

$$\frac{7}{24} - \left(\frac{9}{20}\right)^2 = \frac{107}{1200} > 0.$$

Translations satisfy

$$\text{Ber}(aI + D) = a + \text{Ber}(D), \quad W(aI + D) = a + W(D).$$

For $9/20 < a < r_0$, the first set is the disk of center a and radius $b < a$, hence lies in the sector of semi-angle $\arcsin(b/a) < \pi/2$. But $a - r_0 < 0$ belongs to the second set, so no sector S_θ with $\theta < \pi/2$ can contain it. This proves assertion 3.

Finally, the norm of a unilateral weighted shift is the supremum of its weights. Here $\alpha_2 = \sqrt{2/3}$, while $\alpha_1 < \sqrt{2/3}$ and, for $n \geq 3$,

$$\alpha_n \leq \frac{n}{2^{n-1}} \leq \frac{3}{4} < \sqrt{\frac{2}{3}}.$$

Consequently $\|D\| = \sqrt{2/3} < 1$. Therefore

$$W(I + D) \subseteq \{z : |z - 1| \leq \sqrt{2/3}\},$$

a disk contained in a sector of semi-angle strictly below $\pi/2$; hence $I + D$ is sectorial. Since $\text{Ber}(I + D) = 1 + b\mathbb{D}$, elementary tangent geometry gives $\theta_{\text{Ber}}(I + D) = \arcsin b$. Since $1 + r_0\mathbb{T} \subseteq W(I + D)$, the same tangent geometry gives $\theta_{\text{num}}(I + D) \geq \arcsin r_0$. The strict chain in assertion 4 follows. $\square$

5. EXACT VERIFICATION

Run from the packet directory:

```
conda run --no-capture-output -n sandbox python \
  code/verify_radius_gap.py
```

The script constructs P, N, q from their displayed definitions, converts q to degree-32 Bernstein form, checks all 33 coefficients with `Fractions.Fraction`, confirms the exact minimum displayed above, and checks the exact gap $107/1200$. Its terminal line is

CERTIFIED: `b < 9/20 < sqrt(7/24)`

This is an exact finite certificate, not a floating-point sampling argument. An exploratory decimal estimate $b \approx 0.433904$ was used only to select the convenient rational separator and is not used in the proof.

6. NOVELTY CHECK, LIMITATIONS, AND REVIEW RECOMMENDATION

On 12 August 2026 we searched the run’s registry, solution, attempt, and proof-gap indexes for arXiv:2601.02817 and the phrases *Berezin sectorial*, *composition-differentiation*, and *Dirichlet space*; no duplicate or earlier result was found. A bounded arXiv web search used the source id and the combinations “Berezin sectorial” with “composition-differentiation”, “Dirichlet”, and “different indices”. It found the source paper but no later paper answering this construction question. This is a bounded search rather than an exhaustive MathSciNet, zbMATH, or citation-graph review, so novelty confidence is moderate rather than definitive.

The result is full for the source’s either/or construction request and in fact supplies both alternatives. It does not classify all symbols, compute the exact constants b or $w(D)$, or address unshifted D_ψ without the scalar translations explicitly contemplated by the source. Human review should focus on (i) the Dirichlet normalization, (ii) the Berezin-symbol calculation, (iii) the exact Bernstein conversion, and (iv) agreement that the source problem concerns scalar shifts. Subject to those checks, promotion as a full affirmative answer is recommended.

REFERENCES

- [1] S. Mahapatra, S. Mukherjee, A. Sen, R. Birbonshi, and K. Paul, *An introduction of Berezin sectorial operators and its application to Berezin number inequalities*, arXiv:2601.02817, 2026.
- [2] K. E. Gustafson and D. K. M. Rao, *Numerical Range: The Field of Values of Linear Operators and Matrices*, Springer, New York, 1997.